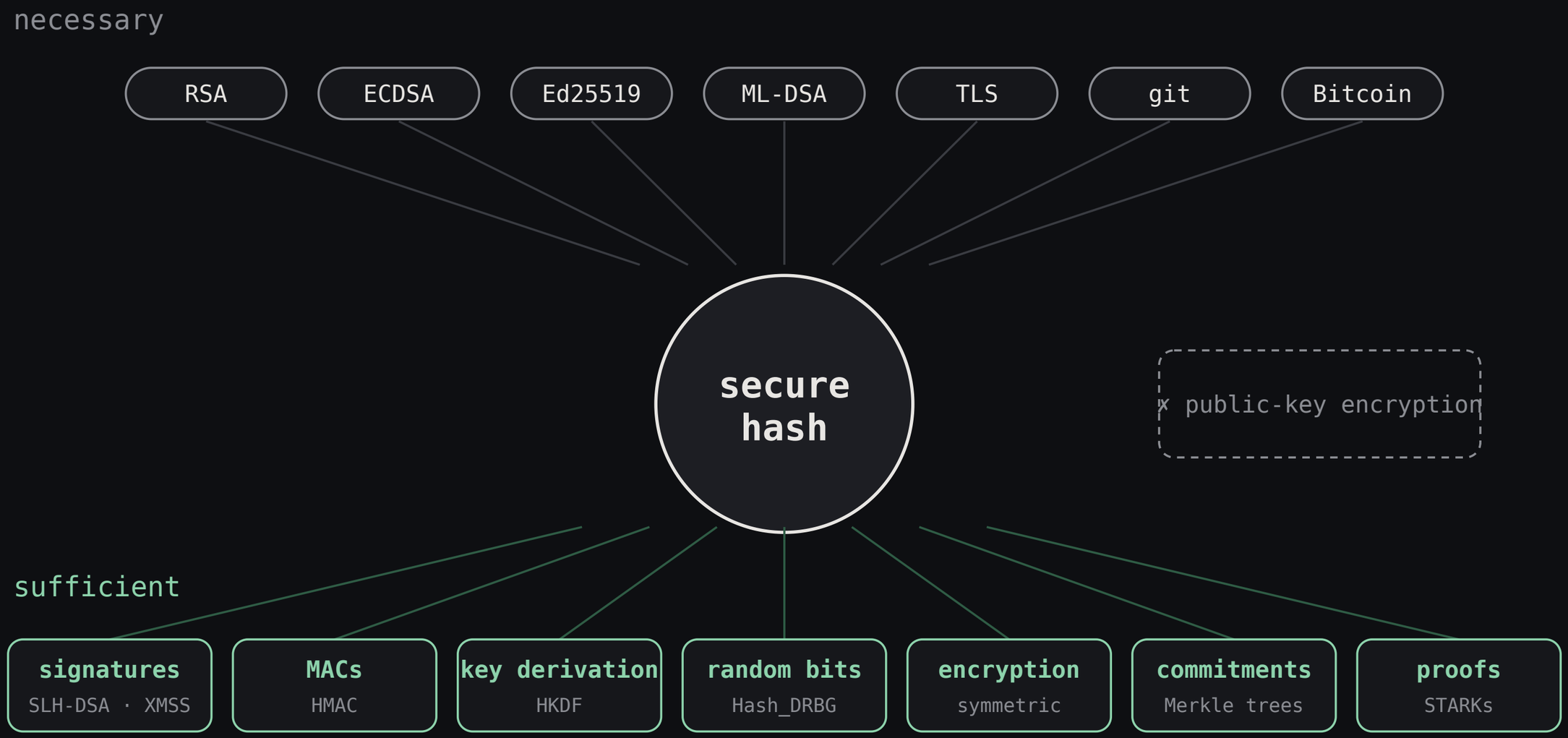
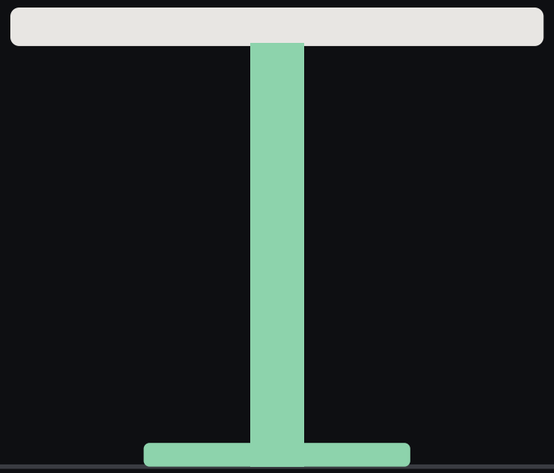


https://HashSma.sh

Updated 2 November 2026, after the Cambridge workshop


necessary
RSA
ECDSA
Ed25519
ML-DSA
TLS
git
Bitcoin
secure hash
✗ public-key encryption
sufficient
signatures
SLH-DSA · XMSS
MACs
HMAC
key derivation
HKDF
random bits
Hash_DRBG
encryption
symmetric
commitments
Merkle trees
proofs
STARKs

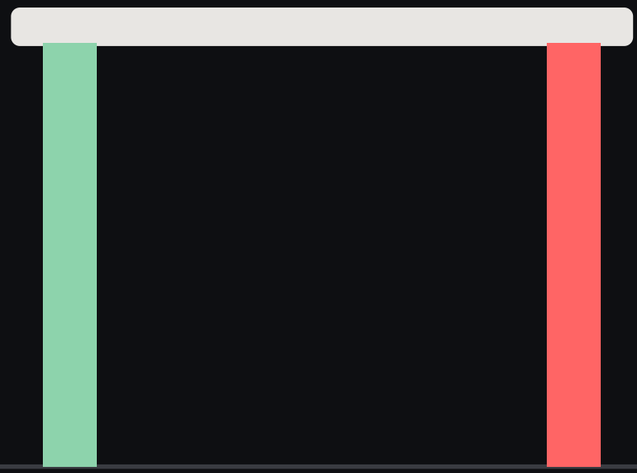
hash-based system



hash

falls if the hash breaks

anything else

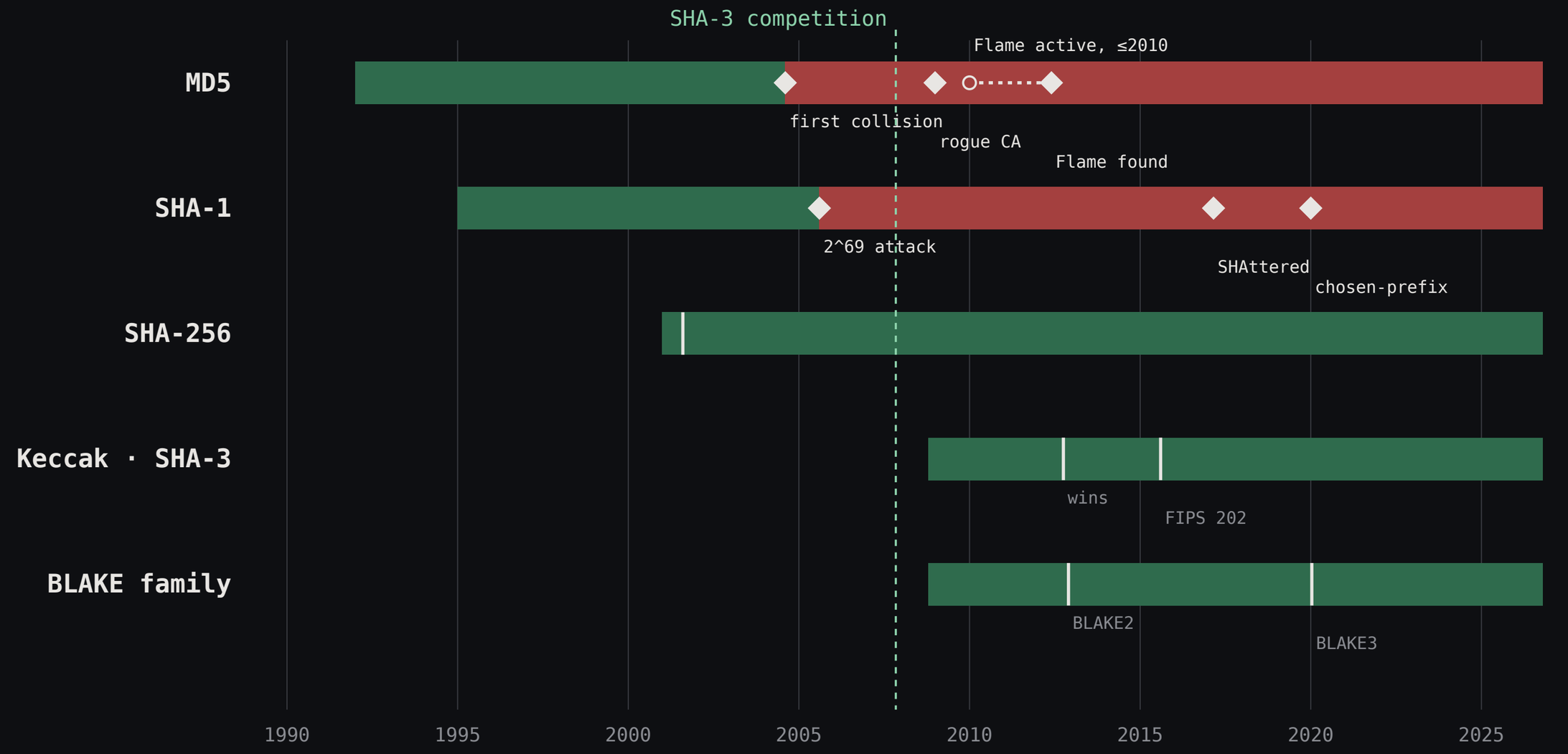


hash **lattices, curves, McEliece, etc.**

falls if the hash breaks
or if the other leg does



After Zooko, “lessons from the history of attacks on secure hash functions” (2017), Figure 0

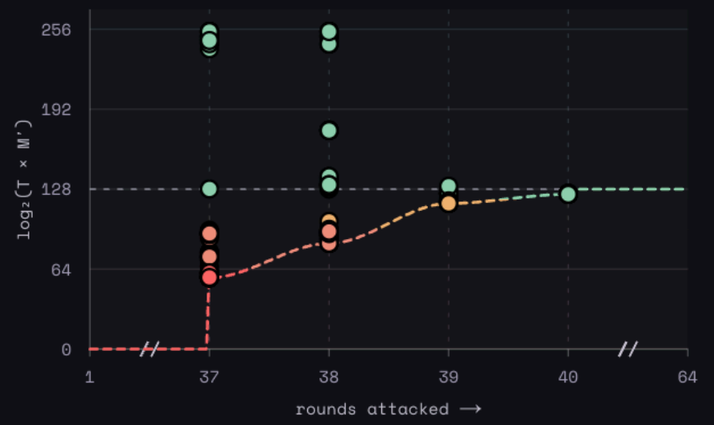


green: no collision attack · red: collisions in practice · ◆ collision milestone

SHA-256

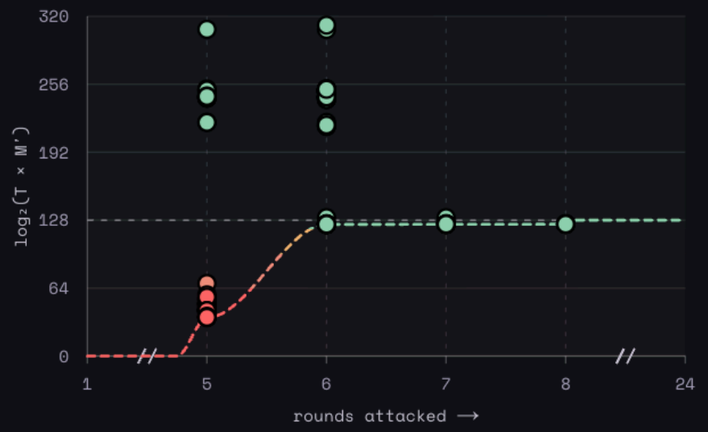
258 submissions

View Results



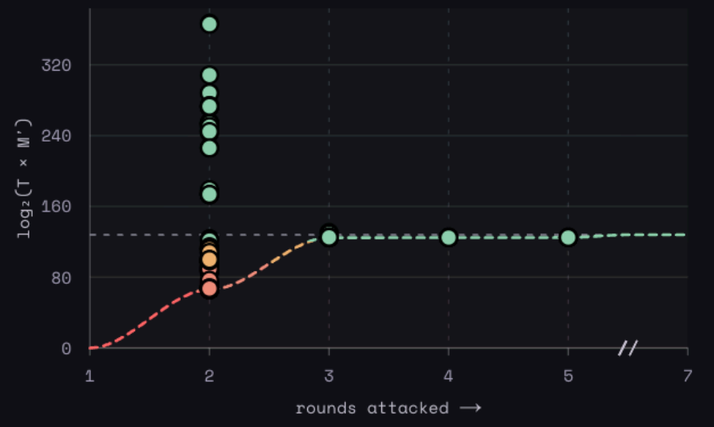
SHA3-256

184 submissions



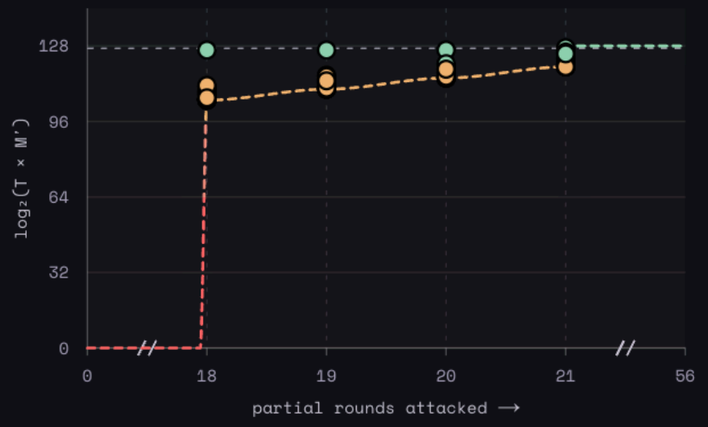
BLAKE3

222 submissions

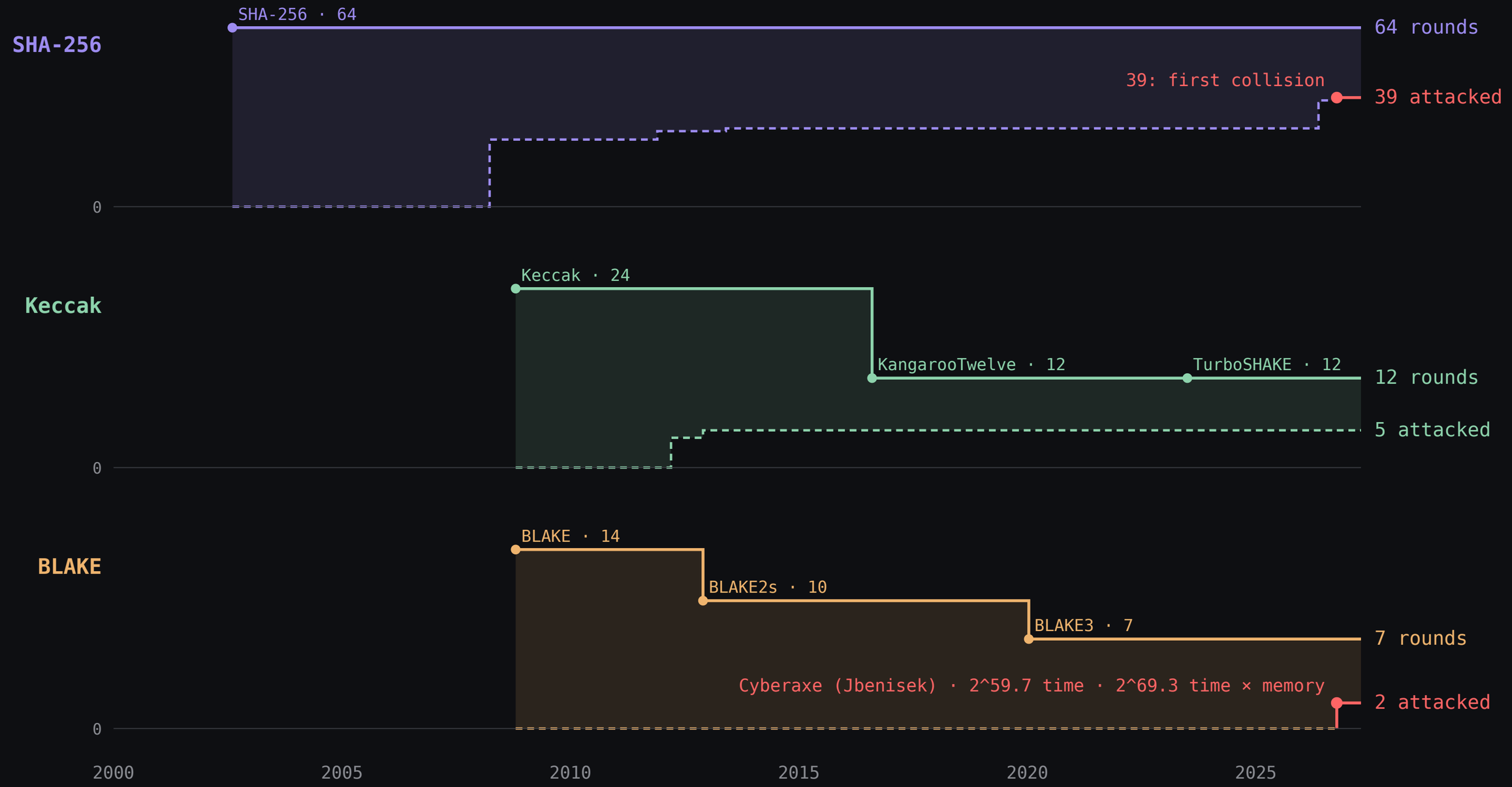


Poseidon (Orchard)

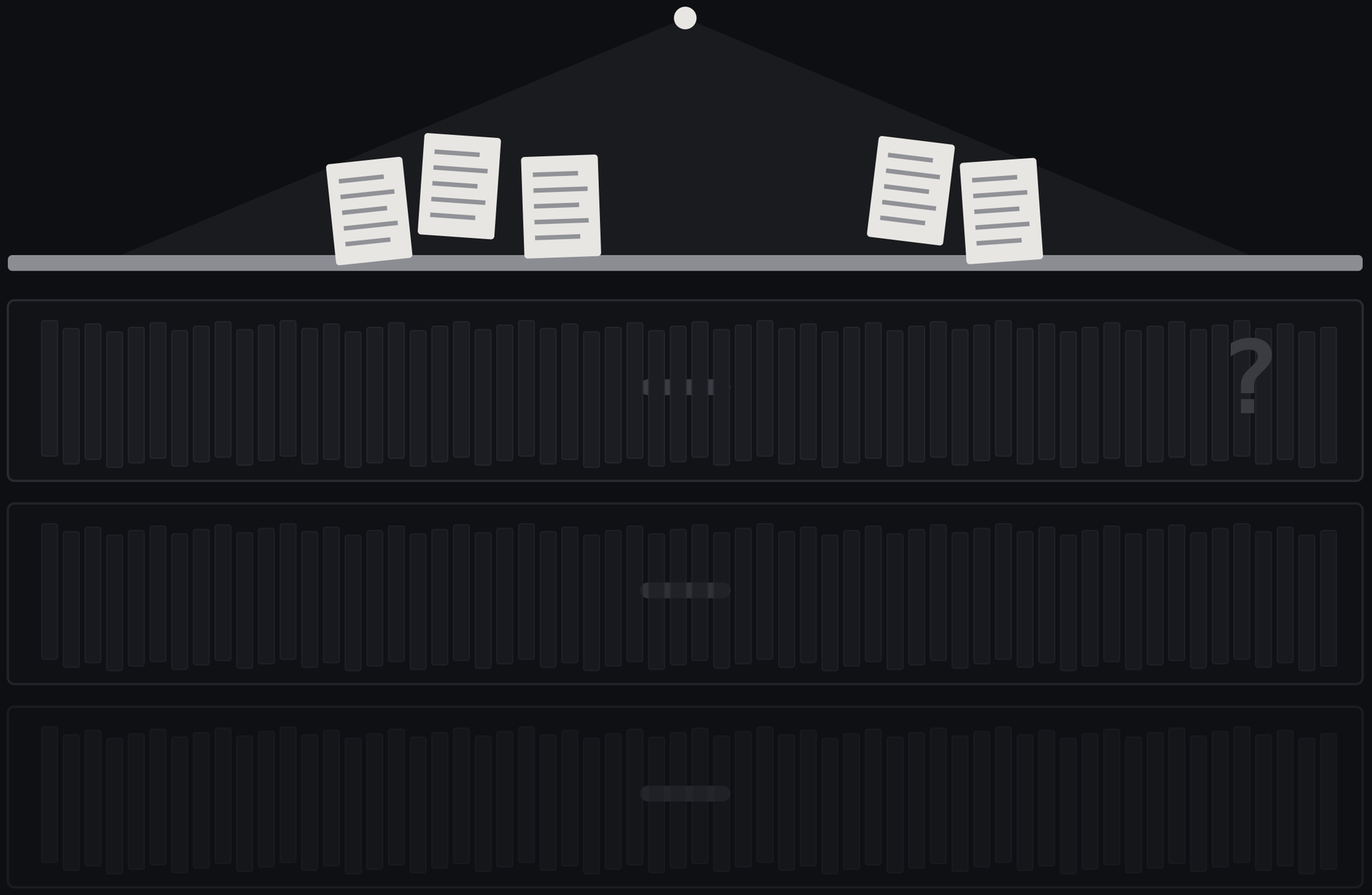
82 submissions



Lowest cost by round · Safe · Wounded · Unsafe · ⚠ VERY Unsafe · M' = max(1, memory / 1 MiB)



solid: rounds the designers chose · dashed: rounds with a published full collision · shaded: the margin · red: HashSmash





Questions

https://HashSma.sh

https://x.com/shieldedlabs

https://x.com/zooko